

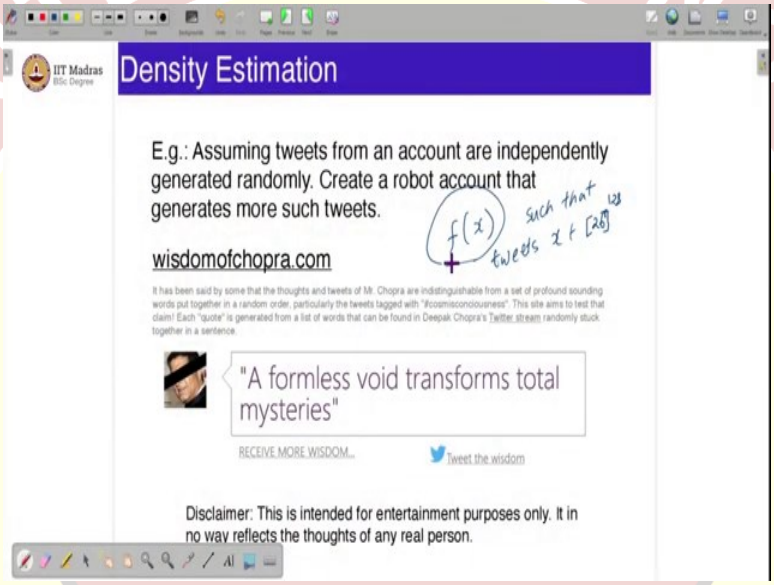
IIT Madras

ONLINE DEGREE

Machine Learning Foundations
Professor. Harish Guruprasad Ramaswamy
Department of Computer Science and Engineering
Indian Institute of Technology, Madras
Unsupervised Learning: Density Estimation

So, with that we wrap up the dimensionality reduction problem and move on to the next unsupervised learning problem, which is the density estimation.

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Density estimation is, the output of a density estimation algorithm is a probabilistic model. Recall that I had mentioned earlier the probabilistic model is a model which essentially scores different configurations of reality. For example, let us say your goal is to build a model, which can generate more tweets, which look very similar to already generated tweets. So, assuming tweets from an account are independently generated randomly, your goal is to create a robot account that generates more such tweets.

Here is such a site built called wisdomofchopra.com, where it has been said by some that the thoughts and tweets of Mister. Chopra are indistinguishable from a set of profound sounding words put together in a random order. So, based on this, they built a probabilistic model, and which generates tweets that look similar to the tweets of Mister. Chopra.

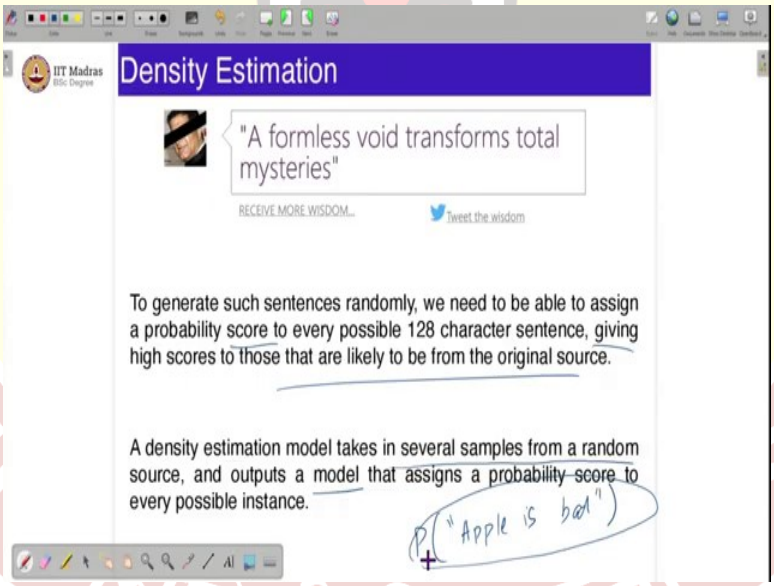
This is an example of how you can use a density estimation or a probabilistic model to generate robots. So, whenever someone wants to generate a robot Twitter account, effectively, a density

estimation is the main tool that is used here. So, what is, what does a density estimation algorithm give you? It gives you a probabilistic model, which scores all tweets, effectively, tweet is something, is a is 128 length array.

So, that is, let us say, for example, a tweet is 128 length character array, which in this case, what you need is a scoring function, which would score any x such that tweets x belonging to 26^{128}. So, I am assuming that you are writing always in lowercase. So, you take there are 26 possible outputs for each space 27 if we include a space and onetwo8 characters, so that is the total number of legal tweets possible.

So, your f of x would essentially take such a thing, and it would say how likely was this tweet, how likely is that this particular tweet can be generated by Mister. Chopra. So, that is how do you get such a, this is a probabilistic model. And constructing such a probabilistic model is the goal of density estimation.

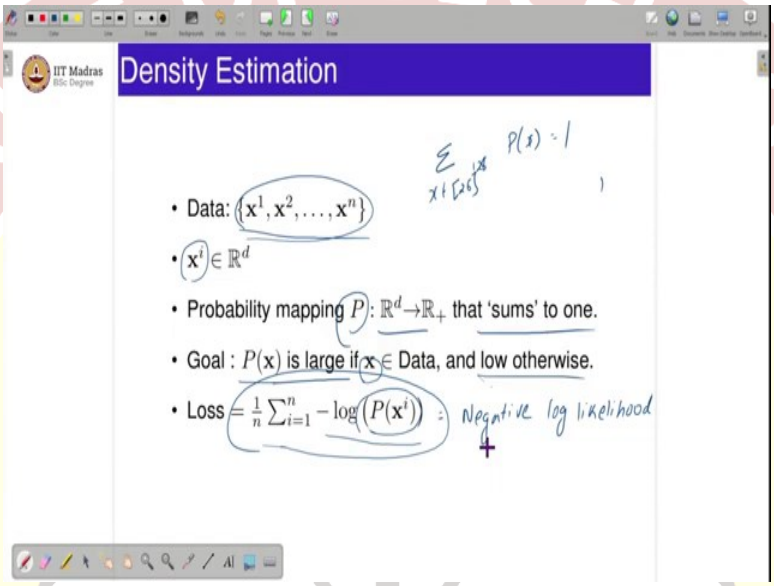
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To generate such sentences randomly, we need to be able to assign a probability score to every possible onetwo8 character sentence giving high scores to those that are likely to be from the original source. How does it do that? Well, a density estimation model takes in several samples from a random source, but in this case, it would take several samples from the original Twitter stream and it will output a model probabilistic model that assigns a probability score to every possible instance. So, by every possible instance, we mean every tweet.

So, we can give, we can say, what is the probability that "Apple is mad"? It should be able to the output of the learning algorithm in the case of a density estimation would simply be a function which would be able to say what is the probability that Apple is bad? So, this is a sentence Apple is bad is one possible output. And you can ask the question, what is the probability that this particular tweet is generated by Mister. Chopra.

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In mathematical terms, once again, your data is it should be a collection of tweets maybe, collection of n tweets, x1 to xn, where each data point is a d dimensional vector. The goal of a density estimation learning algorithm is a probability mapping P, which is a mapping from $\mathbb{R}^d$ to $\mathbb{R}^+$, so it is only positive numbers are allowed. And here is the key thing. It must sum to 1.

That is the reason why you, that would mean that sum over x in 26^{128} probability of x should be equal to 1. So, that you cannot give very high scores to all the tweets, that is just not possible. Because each tweet has to have a positive score or 0 score, but the total probability of, the total score has to be equal to 1. So, that is the goal, the goal of a density estimation algorithm is to come up with such a probability model.

And what property models are desirable? What is the last function for this? The goal is that if P of x is large, if your x belongs to this data, and it should be low otherwise, that is the goal here. And of course, it can be large. Otherwise, also, if the data is some x is very similar to an already existing

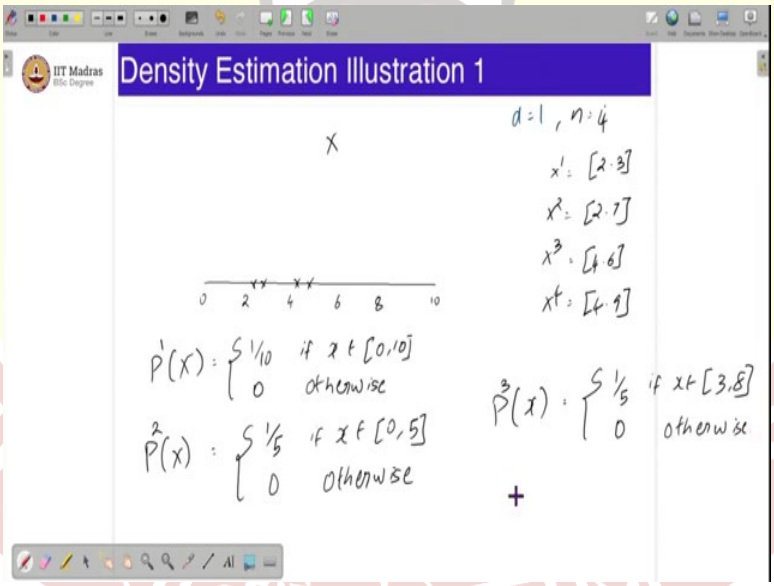
data. So, the last that is used is called the negative log likelihood, which is simply defined as follows.

For each data point, xi, you want your P of xi to be as large as possible. And in other words, you want your negative log of P of xi to be as small as possible. What is the ideal case scenario, the ideal case scenario would be you would have log of, P of xi would be equal to 1 by m, does not matter.

So, P of xi would be very high for x1 to, P of x1 would be high, P of x2, P of x3 would be high, P of xn would all of that would be high. So, that - log of P of xi would be small. And when you sum this over an average over all the n data points, this would be small. So, this is the average negative log likelihood. Your goal of a density estimation algorithm is to come up with a probability model or a probability mapping P, which has a small negative log likelihood as possible.

Here is have a couple of examples of how to do that. Here is a very simple illustration problem.

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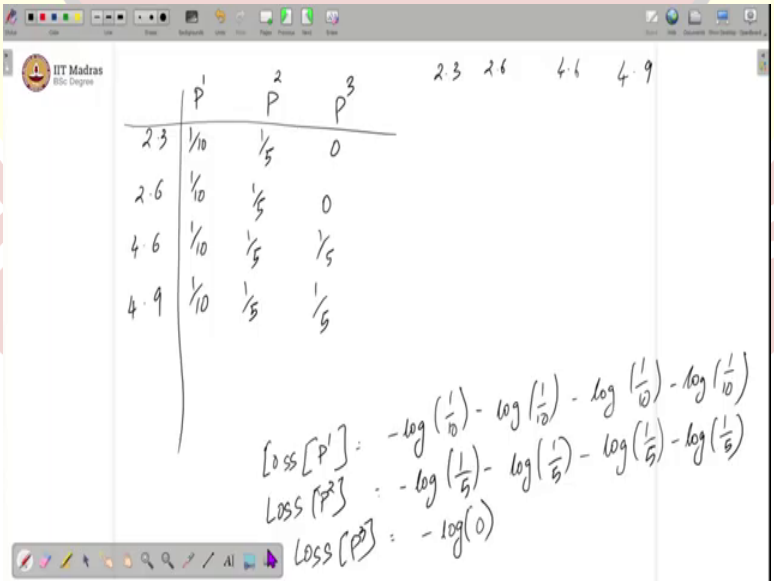
Let us say you have your, in this case, D = 1, you have just one-dimensional data. And let us say your data, you have 4 data points, this n equals 4 the data points are given by, let us say, x 1 = let us say 2.3, x, sorry about the superscript should be a superscript. Say x1 = 2.3, x2 = let us say 2.7, x3 = let us say 4.6, x4 is let us say equal to 4.9.

So, you have 4 data points, let us maybe plot that as on the number line, from 0 to 10, you have 2.3, and 2.7. And you also have 4.6, and 4.9, you have these 4 data points. Now, once again, in the spirit of things that we have done so far, we will give a bunch of models, and we will ask the density estimation algorithm to choose the best model out of these, so what are the models here?

So, the first model we can consider is that P of x = 1 by 10, if x belongs to 0 to 10, and 0 otherwise. So, I forgot to mention so in this case, you are, it is not a discrete thing like a tweet where you can take only 1 of 26 possible values. So, here your X can take a real valued thing. X is a real valued instance. So, summing to 1 is essentially replaced by integrating to 1, so that is why I have 1 by 10, 0 to 10. So, if you integrate this function over the rail line, you will get the value would be equal to 1. So, that is why I am using 1 by 10. So, this is one possible probability model.

Let us have another probability model P of x =, let us say, 1 by 5, if x belongs to 0 to 5, and 0 otherwise, this is also a valid probability model. So, then we will use, finally we will use another model called P1, P2 and let us call this P3. P3 of x = 1 by 5. If x belongs to let us say 3 , 8, x is between 3 and eight, it is 1 by 5 and 0 otherwise. All these three are valid probability models. Now, I just want the learning algorithm to choose between these three models. Which of these three has the lowest negative log likelihood? So, if you could compute that. Let us do that.

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Let us recall the data once again, it is 2.3, 2.6, 4.6, 4.9. What would the scores of each of these components 2.3, 2.6, 4.6, 4.9. According to P1, P1 is just uniform between 0 to 10, so it would

give a score of 1 by 10, 1 by 10, 1 by 10, 1 by 10. For all of these 4 points. P2 is uniform within 0 to 5, it will give a score of 1 by 5, 1 by 5, 1 by 5, 1 by 5.

P3 is uniform in the range of 3 to 8. Which means that, according to P3, the score for 2.3 and 2.6 should actually be 0. And 1 by 5, 1 by 5, we can compute the negative log likelihood that is loss of the model P1 would be - log 1 by 10 - log 1 by 10 - log 1 by 10 - log 1 by 10. So, this would be the loss of the probabilistic model P1. What would be the loss of P2?

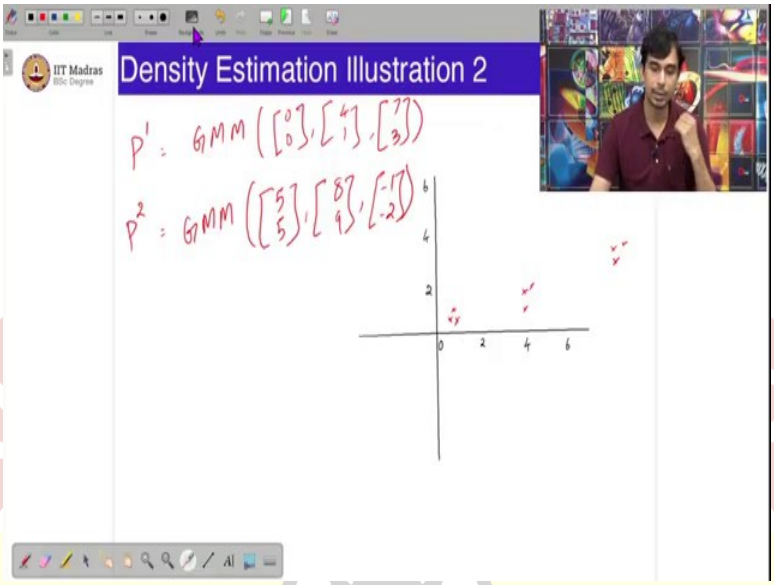
That would be - log 1 by 5, - log 1 by 5, - log 1 by 5, - log 1 by 5. What would be the loss of P3? Here is, I think here you would have - log of 0, it actually does not matter what you write for the rest of the things because - log of 0 is already infinity, log of 0 is - infinity and - log of 0 is infinity. So, the loss of the model P3 is going to be infinity, regardless of how the other things are going to be.

So, you can clearly see that loss of P1 and loss of P2 are going to be not infinity. So, which is better, we can clearly see once again that loss of P2 is going to be smaller than loss of P1. So, P2 is the better probabilistic model for explaining this data than P1. And it seems logical, P1 says that these 4 data points are coming from uniformly between 0 to 10.

P2 is saying that these data points are coming uniformly between 0 to 5. P2 seems to be a better model than P1 for this particular data. Of course, I am not saying that is the best model in the context of this particular type problem P2 is better than P1 because P2 has lower negative log likelihood than P1.

So, in a real density estimation model would not once again be given multiple choice questions like this, it would not be given P1, P2 and P3, you would actually be, it would actually find the best probabilistic model from an infinite set of models. So, that is one example. Let us move on to another example.

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So, here is another example. Let us say you have 9 points. So, here is a slightly more complicated class of models, which you can estimate. Let us say you have 9 data points, I am just finding and I am not going to write down the 9 points, here these are the 9 points. I have put an X in the Cartesian plane for illustration these 9 points, so you can have two models now.

One model is a Gaussian mixture model which says that the data is centered around 3 P1 is a, I am not going to explain what the Gaussian mixture model is right now, but broadly wanted to saying is, the Gaussian mixture model centered at 0 , 0 and 4 , 1 and let us say 7 , 3. That is one Gaussian mixture model. And there is another Gaussian mixture model, which says that this is data centered around 3 possible centers, just let us say 5 , 5, 8 , 9, - 1 , - 2, which is a better model?

I am not going to go into the numerical details, but you can immediately see that out of these two models, you have, you can look at the data and you can see that it seems to have 3 centers, one center around here and one center around here and another center around here. So, based on this P1 is a much better explanation for this data than P2.

So, you are, if you had actually computed the negative log likelihood of these two models P1 and P2 and chosen the model with the lesser negative log likelihood, you would have chosen P1. Once again real density estimation models do not have multiple choice questions, but rather they are given entire parametric family of models and they choose the model among this infinite models, which has the least negative log likelihood or which has the least loss in this particular case.

This is once again there are several extensions possible to this based on here in the previous case, we had uniform models that are uniform, here in this case, we had the models were all Gaussian mixture models. You can have several other examples in the same way that you had for regression and classification and dimensionality reduction. You can have similar other examples for density estimation also.

And with that, I think you have a broad overview of the two main unsupervised learning problems, which are dimensionality reduction and density estimation. With that will wrap up the introductory week of the Machine Learning Foundations course. And later on in the course, we will dig deeper into all of these learning problems and actually come up with ways to formulate the learning algorithm in the full power that is to find the best model in the entire model, in the entire class of models, in the entire library of models.

Which is how do you write, how do you construct learning algorithms, which finds the best model among an infinite family of models and that we will cover later in the course when we do individual problems which are highlighted as a special cases of the tools that you will learn throughout the course. That will wrap up the first week. Thank you.